Certificates as Constraints: Feasibility-Preserving Multi-Agent Reinforcement Learning for UAV Swarms Whose Guarantees Degrade

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Abstract

Safe reinforcement learning offers two kinds of guarantee, and neither fits a fleet whose safety margins come from a formal certificate. Lagrangian methods satisfy a constraint *in expectation*. Shields and barrier functions hold it at every step, but against a safe set fixed in advance. We study allocation in an unmanned aerial vehicle (UAV) swarm in which each aircraft carries a certified protection level $\rho_i(t)$: a sound bound on its position error that grows under global navigation satellite system (GNSS) jamming and jumps under undetected relay-delay attacks. A task with alert limit α_j may be held only while $\rho_i(t) \leq \alpha_j$. The feasible set therefore moves with the threat the policy is reacting to. We make three contributions. First, a projection layer placed inside the environment makes every executed assignment feasible at every decision epoch, for any policy, during training and deployment. It also covers the maximum number of tasks, and policy gradients stay unbiased for the projected process. Second, because a scalar certificate makes eligibility nested, classical matching theory yields exact, linear-time monitors: certified coverage, the number of certificate collapses the fleet tolerates, and the exact worst case under graded degradation. We add a hysteresis bound and a predictive rule under which inter-epoch exposure comes only from jumps. With measured constants, that rule needs an allocator rate of 5.01 Hz before every aircraft at its benign level can hold a 2 m corridor. Third, a population lift turns the per-operation guarantee into fleet-sizing statements and lists its independence assumption as an evidence obligation. When common-cause jammers violate that assumption, the independent model errs in both directions. It understates the probability of full coverage by $3.4\times$ and the probability of losing six of eight tasks by $3.7\times$. Every constant is derived from released measurements (15,392 relay hops, 674 GNSS-degradation samples). A brute-force oracle confirms every closed form on 83,872 checks with 0 disagreements. In a multi-agent study, the layer executes no infeasible assignment during learning, where a Lagrangian baseline executes 3,903,179. Added at deployment, it lifts every learned policy to maximal coverage.

1 Introduction

A swarm of unmanned aerial vehicles (UAVs) flies a monitoring or delivery mission. Each aircraft's navigation stack now carries a formal certificate. A recent line of work bounds how far an aircraft can drift when an adversary delays its navigation input just below an intrusion detector's threshold [14]. Its cryptographic cost is quantified in [12], and its relation to aviation safety objectives in [13]. The certificate is not a constant. It grows while a jammer forces the aircraft to dead-reckon, and it jumps when a compromised relay delivers a stale correction. Each mission task, in turn, tolerates only a bounded position error: a 2 m corridor between buildings, a 20 m transect over open ground. The operator's question is therefore dynamic: *which aircraft may hold which task, now, given what the certificates say?*

This is a constrained sequential decision problem, and the safe reinforcement learning (RL) literature has the pieces but not the composition. Constrained Markov decision processes (CMDPs) and

their Lagrangian and trust-region solvers, such as constrained policy optimisation (CPO), proportional–integral–derivative (PID) Lagrangian methods and multi-agent CPO (MACPO) [4, 10, 31, 74], satisfy the constraint *in expectation over the horizon*. A fleet that flies outside a certified envelope on 5% of steps but satisfies the budget on average is not acceptable. Expectation is the wrong quantifier for flight. Shields, safety layers and control barrier functions (CBFs) [9, 11, 25, 27] hold the constraint at every step, but against a safe set given by a specification or by known dynamics. Their guarantee concerns safety, not mission utility, even in the dynamic and time-varying variants. Certified RL [48, 79, 82] certifies one policy against a fixed perturbation ball and never feeds a certificate into a fleet-level constraint. What is missing is an analysis in which *the constraint set of a multi-agent policy is the output of a formal certificate that moves adversarially*.

What is and is not new. We state the boundary at the outset, because the combinatorial core is classical. With a scalar certificate, eligibility has nested neighbourhoods: an aircraft that qualifies for a demanding task qualifies for every easier one. On such *difference graphs* [34], Ore's deficiency theorem [61] gives the maximum coverage in closed form. The condition for surviving b removals is the b-extended Hall condition from robust assignment [49]. Pointwise feasibility under projection is the standard guarantee of shielding [9, 46]. We use these as tools, with attribution. The contribution is their composition with a certificate that moves adversarially, and four results that composition makes possible.

Contributions.

1. **Certificate-as-constraint formulation** (Section 2). The aviation integrity rule "protection level below alert limit" becomes a pointwise CMDP constraint over an exogenous, adversarial, càdlàg process (right-continuous with left limits). We grade the guarantee by quantifier (Table 1) and identify what distinguishes it from state-wise, almost-sure and anytime constraints [57, 72, 88].
2. **A certified projection layer** (Theorem 1). Placed in the environment, it makes every executed assignment feasible at every decision epoch for any policy parameters. It assigns the maximum number of tasks, then maximises assigned value, then agrees with the policy wherever it can. Under a Markov threat model the projected process is itself a Markov decision process, so learning sees no additional projection bias. This mitigates the aliasing problem of projected RL [30, 56]. A greedy variant runs in $O(N \log N + M \log M)$ and handles 10^5 aircraft in $40\,\text{ms}$.
3. **Exact monitors and dynamics** (Section 3):
   - worst-case coverage after any b certificate collapses, and after graded degradation of any b aircraft; the graded bound is exact and attained by an adversary (Propositions 1 and 2);
   - a hysteresis bound on forced releases in terms of the certificate's positive variation (Theorem 2);
   - a predictive rule under which exposure between decision epochs comes only from certificate jumps, with the allocator rate it implies (Theorem 3);
   - a misreporting bound (Proposition 3).
4. **A population lift with explicit obligations** (Proposition 4). It turns the per-operation guarantee into fleet-sizing and tail statements, addressing the gap between per-operation and per-population quantifiers identified by the companion systematisation [13]. Experiments show what happens when its independence obligation fails.
5. **A measured, reproducible study.** Every constant is derived from released flight and network data. A brute-force oracle verifies every closed form. A multi-agent reinforcement learning (MARL) study and a scalability study answer six research questions (RQ1–RQ6). They separate what a learned policy contributes from what the mechanism enforces: the mechanism supplies the guarantee and almost all of the coverage, and learning adds a modest gain on top.

2 Preliminaries and Problem Statement

2.1 The certified protection level

The parent certificate [14] considers an adversary that delays relayed navigation messages while staying below a detector's threshold. Let $\Delta_i(t)$ be the *staleness* of aircraft i's navigation input: the age of the newest trustworthy fix it is acting on. Staleness grows at one second per second while the aircraft dead-reckons under jamming or a sensor fault. It jumps by the hidden per-hop residual when a compromised relay delivers a delayed correction. Against this adversary, the certificate bounds the position error by a Grönwall tube,

$$\|\hat{p}_i(t) - p_i(t)\| \;\le\; \rho_i(t) \;\coloneqq\; e^{LT}\,\gamma_{\sup}\,\Delta_i(t), \tag{1}$$

where $\gamma_{\sup}$ is the supremum rate at which staleness becomes position error and e^{LT} amplifies it through closed-loop dynamics with local Lipschitz constant L over a horizon T. All three factors are measured in the released artifact:

- $\gamma_{\sup} = 1.625$ m/s over 674 post-onset samples of real Global Positioning System (GPS) jamming and spoofing flights;
- $L = 1.181$ with $T = 1$ s;
- benign residuals no larger than 0.178 s, and undetected malicious residuals with median 3.94 s and maximum 4.84 s, over 15,392 relay hops.

Writing $L_\rho \coloneqq e^{LT}\gamma_{\sup} = 5.30$ m/s, a jammed aircraft's protection level grows at most at rate L_ρ. A single undetected two-hop relay attack moves it by about 41.8 m, past every alert limit we use. The benign floor is $\rho_0 = L_\rho \cdot 0.178 \approx 0.94$ m.

We borrow two terms from GNSS integrity monitoring. The *protection level* is a runtime bound on position error. The *alert limit* is the largest error an operation tolerates. An operation is unavailable whenever the protection level exceeds the alert limit [1]. We call $\rho_i(t)$ the *certified protection level*. Each task j carries an *alert limit* $\alpha_j > 0$, its corridor half-width ($\{2,5,10,20\}$ m in our missions). Unlike a receiver protection level, which overbounds error at a target integrity risk, (1) is a worst-case bound conditional on stated assumptions [14]. The difference is the per-input versus per-population gap that [13] systematises, and Section 3.4 returns to it.

Assumption 1 (Certificate soundness, from [14])**.** Whenever $\rho_i(t)$ is reported, $\|\hat{p}_i(t) - p_i(t)\| \le \rho_i(t)$. This work consumes $\rho_i(\cdot)$ and makes no assumption about how it is computed. We model $\rho_i(\cdot)$ as an arbitrary nonnegative, bounded càdlàg process (right-continuous with left limits), chosen by an adversary: a jammer is not a distribution. Only Theorem 1(iii) and the learning study additionally assume a Markov threat model.

2.2 The fleet constrained Markov decision process

A fleet $\mathcal{N} = \{1, \dots, N\}$ serves tasks $\mathcal{T} = \{1, \dots, M\}$, each with a site, an alert limit α_j and a value $v_j > 0$. Decisions are taken at epochs $t_k = k\Delta t$. An *assignment* is a map $\sigma : \mathcal{N} \to \mathcal{T} \cup \{\varnothing\}$ that is injective on $\mathcal{T}$, so each task has at most one holder; it is a matching in $\mathcal{N} \times \mathcal{T}$. The fleet is a CMDP [10] whose state s_t contains positions, task sites and every $\rho_i(t)$. Its action is an assignment, and its reward is *certified coverage*: the value of tasks whose holder is on site and certified, $r_{\mathrm{cert}}(t) = \sum_j v_j \mathbb{1}[\text{holder of } j \text{ on site and } \rho_{\sigma^{-1}(j)}(t) \le \alpha_j] / \sum_j v_j$. The constraint is pointwise:

$$\text{(C)} \qquad \rho_i(t_k) \;\le\; \alpha_{\sigma_{t_k}(i)} \qquad \text{for every } i \text{ with } \sigma_{t_k}(i) \neq \varnothing, \text{ every epoch } k, \text{ every episode.} \tag{2}$$

The *eligibility graph* $G(t)$ joins i to j iff $\rho_i(t) \le \alpha_j$, and (C) says that executed assignments are matchings of $G(t_k)$. Sort the alert limits ascending, $\alpha_{(1)} \le \cdots \le \alpha_{(M)}$, so the most demanding task comes first, and define the *count profile* $n_k(t) \coloneqq |\{i : \rho_i(t) \le \alpha_{(k)}\}|$. Since $\rho_i \le \alpha_{(k)}$ implies $\rho_i \le \alpha_{(k')}$ for all $k' \ge k$, neighbourhoods in $G(t)$ are nested. $G(t)$ is a *difference* (chain) graph [34, 54].

Table 1: Quantifiers on a learned constraint. "Exogenous" means the constraint set is driven by a process the agent does not control.

Quantifier	**Representative methods**	**What is guaranteed**
In expectation	CPO, PPO-/PID-Lagrangian, MACPO [4, 31, 74]	$\mathbb{E}[\sum_t \gamma^t c_t] \leq d$, possibly after training
High probability, state-wise	state-wise CPO (SCPO) and variants [88, 89]	per-state constraint with probability $1 - \delta$
Almost sure, own budget	Sauté RL, anytime-constrained RL [57, 72]	the agent's own cumulative cost never exceeds its budget
Every step, specified safe set	shields, safety layers, CBFs [9, 11, 25, 27]	state stays in a set given by a specification or known dynamics; no utility guarantee
Every epoch, certificate-driven set	**this work**	**(C) at every epoch for a set moved by an adversarial certificate; coverage maximal**

2.3 Problem statement

Given a policy class that proposes assignments from observations, we seek four things:

P1 an allocation mechanism under which (C) holds at every epoch of training and deployment, for any policy parameters, without collapsing coverage. Leaving every aircraft idle satisfies (C) trivially, so coverage must be part of the guarantee;

P2 an exact, real-time characterisation of the coverage the fleet can certify, and of how it degrades when certificates collapse or grow;

P3 rules that keep the guarantee meaningful between decision epochs and avoid chattering as $\rho_i(t)$ fluctuates;

P4 a sound route from the per-operation guarantee ($\mathbf{Q}_{\mathrm{op}}$) to the per-population statements that fleet sizing and airworthiness objectives require ($\mathbf{Q}_{\mathrm{pop}}$), with every assumption made explicit.

Figure 1 summarises how the paper addresses P1–P4. Section 3 develops the theory and Section 4 the experimental method, and Section 5 answers six research questions:

- **RQ1**: zero violations during training versus a Lagrangian baseline;
- **RQ2**: coverage as certificates degrade, against the bounds;
- **RQ3**: comparison with baselines;
- **RQ4**: scalability;
- **RQ5**: dynamics between decision epochs;
- **RQ6**: the population lift.

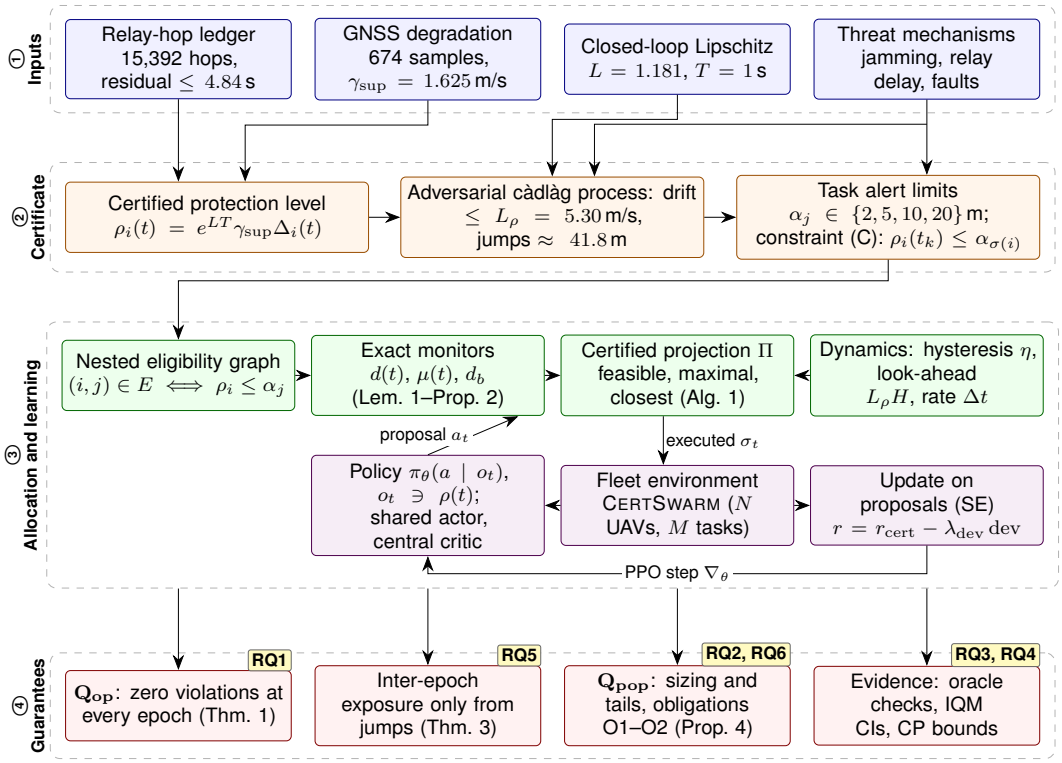


Figure 1: Methodological framework, read top to bottom. ① Measured quantities from the parent artifact determine ② each aircraft's certified protection level $\rho_i(t)$, a bounded càdlàg process that drifts under jamming and jumps under relay delay; a task may be held only while $\rho_i(t)$ does not exceed its alert limit α_j. ③ Because the constraint is a scalar threshold on each side, the eligibility graph is nested. Closed forms monitor the coverage the fleet can certify, and a projection inside the environment maps every proposal of the learning policy to a feasible, coverage-maximal assignment, so training never executes an infeasible one. ④ The outputs are a per-operation guarantee and a population statement whose assumptions are listed as evidence obligations. Tags give the research question that each output answers.

3 Methodology I: Certified Allocation under Nested Constraints

Proofs are in Appendix A. Throughout, $d(t)$, $n_k(t)$ and $G(t)$ are evaluated at the current certificates. We drop t when no confusion arises.

3.1 Exact monitors

Lemma 1 (Certified coverage; Ore [61] on nested graphs)**.** *The maximum number of tasks coverable by a feasible assignment is $M - d$, where*

$$d \;=\; \max_{1\le k\le M} \big(k - n_k\big)^+. \tag{3}$$

It is computable in $O(N \log N + M \log M)$, and a maximum feasible assignment is produced by processing tasks in ascending α and giving each one any *still-free eligible aircraft.*

Lemma 2 (Coverage margin; b-extended Hall [49])**.** *Full coverage survives the collapse ($\rho \to \infty$) of* any *b aircraft iff $n_k \ge k + b$ for all k. The largest such b is the* coverage margin *$\mu = \min_k(n_k - k)$, and full coverage currently fails iff $\mu < 0$.*

Algorithm 1 Certified projection Π (executed at every decision epoch)

Require: proposal $a \in (\mathcal{T} \cup \{\varnothing\})^N$, certificates ρ, alert limits α, values v, previous assignment σ^-
Require: band $\eta \in [0, 1)$, $h \geq 0$; look-aheads $\ell_{\text{acq}}, \ell_{\text{keep}} \geq 0$
$E_{ij} \leftarrow \big[\rho_i + \ell_{\text{acq}} \leq (1-\eta)\alpha_j - h\big]$ for $j \neq \sigma^-(i)$; $\quad E_{i\sigma^-(i)} \leftarrow \big[\rho_i + \ell_{\text{keep}} \leq \alpha_{\sigma^-(i)}\big]$ ▷ acquire / keep
$W_{ij} \leftarrow B + c_v v_j + \mathbb{1}[a_i = j]$ on E ▷ $c_v = (N+1)/\min \Delta v$, $B > M(c_v \max v + 1)$
$\sigma \leftarrow$ maximum-weight matching of (E, W) ▷ Hungarian; greedy nested variant: Lemma 1
return σ ▷ executed; the policy is trained on a (SE placement)

The margin is a procurement quantity. It counts the simultaneous certificate failures that the fleet absorbs with no loss of coverage. The next two results give the exact worst case when that budget is exceeded.

Proposition 1 (Optimal collapse adversary)**.** *If an adversary may collapse any b certificates, the maximum certified coverage afterwards is $M - d_b$ with $d_b = \max_k \big(k - (n_k - b)^+\big)^+$. The minimum is attained by collapsing the b aircraft with the smallest ρ.*

Proposition 2 (Graded degradation, exact worst case)**.** *Let any b aircraft have their protection level increased by $s \geq 0$, for example by s/L_ρ seconds of dead reckoning. Let $n_k^{(s)} = |\{i : \rho_i + s \leq \alpha_{(k)}\}|$. Then the maximum certified coverage afterwards is at least*

$$M - \max_k \big(k - \max\{\, n_k - b,\ n_k^{(s)}\}\big)^+, \tag{4}$$

and an adversary choosing the b aircraft attains this value. As $s \to \infty$ it reduces to Proposition 1.

The generic bound "each failure costs at most one task", $M - d - b$, is valid but loose. Propositions 1 and 2 are exact, and on our fleets the gap is several tasks (Section 5.4). Classical coverage control treats failed agents as binary [24]. Resilient multi-robot planning bounds submodular utility after α worst-case removals [90, 91]. Proposition 2 is exact for graded degradation, which is what jamming produces.

3.2 The certified projection layer

The projection sits in the environment: the policy proposes, the environment executes Π of the proposal, and the policy is trained on what it proposed. Following Markgraf et al. [56], we call this the *safeguard-in-environment* (SE) placement, as opposed to placing the safeguard inside the policy.

Theorem 1 (Certified projection)**.** *Let Π be Algorithm 1 with $\eta = h = \ell_{\text{acq}} = \ell_{\text{keep}} = 0$. For every policy π_θ, every parameter value θ and every certificate trajectory permitted by Assumption 1, the executed assignment $\sigma_{t_k} = \Pi(a_{t_k}; s_{t_k})$ has the following properties:*

(i) it satisfies (C) at every epoch of every episode, in training and in deployment;

(ii) it assigns exactly $M - d(t_k)$ tasks, the maximum. Among maximum assignments it maximises assigned value, where $\min \Delta v$ is the smallest gap between distinct task values; among those it maximises agreement with the proposal a_{t_k};

(iii) if, in addition, the threat process is Markov in a state s that includes σ^-, the process seen by the policy is a Markov decision process (MDP) with kernel $P_\Pi(s' \mid s, a) = P(s' \mid s, \Pi(a; s))$. Its observations are partial when the policy is decentralised. The likelihood-ratio policy gradient of the projected objective is therefore unbiased, and PPO incurs no bias beyond its usual clipping and critic bias. Differentiating through Π instead gives zero gradient almost everywhere, because Π is piecewise constant in a.

For the coverage-only objective, the greedy nested variant of Lemma 1 achieves (i) and the cardinality part of (ii) in $O(N \log N + M \log M)$.

Two remarks place Theorem 1. First, part (i) alone is easy: the constraint is separable across aircraft, so per-agent action masking [18, 39] also never violates it. Theorem 1(i) is then the familiar shield guarantee, transposed to a certificate-driven set. What masking cannot provide is (ii). Agents that each pick a feasible task still collide, leaving coverage below $M - d$; Section 5.3 measures the difference. Second, the placement in (iii) follows the safeguard-in-environment analysis of [56]. That analysis shows that action aliasing, where many proposals map to one executed action, harms policy gradients more when the safeguard sits in the policy. Our setting has a further adversarial component: under Assumption 1 alone, the projected process is a game against the jammer rather than an MDP, and (i)–(ii) still hold while (iii) does not. We add a small penalty λ_{dev} on the fraction of proposals the projection overrides, so the policy learns to propose assignments the projection keeps.

3.3 Dynamics between and across epochs

Constraint (C) binds at decision epochs. Between epochs the certificate keeps moving, and across epochs an aircraft whose ρ_i hovers near α_j could acquire and drop a task repeatedly. We call a holder *exposed* on $[t_k, t_{k+1})$ if (C) held at t_k and fails before t_{k+1}. A *forced release* occurs when a holder becomes ineligible for its own task. Let $\mathrm{TV}_i^+[0,\tau]$ be the positive variation of ρ_i on $[0,\tau]$: the sum of all its increases, including jumps.

Theorem 2 (Hysteresis)**.** *With band (η, h) in Algorithm 1 and $\ell_{\mathrm{keep}} = 0$, the number of forced releases of aircraft i on $[0,\tau]$ is at most $1 + \mathrm{TV}_i^+[0,\tau]/(\eta\alpha_{(1)} + h)$. If $\ell_{\mathrm{keep}} \le \ell_{\mathrm{acq}} + h$, the number of assigned feasible tasks is at least $M - d(\tilde{\rho})$, where $\tilde{\rho}_i = (\rho_i + h + \ell_{\mathrm{acq}})/(1-\eta)$.*

Theorem 3 (Predictive feasibility and the allocator rate)**.** *Suppose that between upward jumps every ρ_i grows at rate at most L_ρ. If $\ell_{\mathrm{keep}} \ge L_\rho \Delta t$ and $\ell_{\mathrm{acq}} \ge L_\rho H$ with $H \ge \Delta t$, then:*

(i) a holder can be exposed on $[t_k, t_{k+1})$ only if its certificate jumps upward in that interval, so the number of exposed holder-epochs is at most the number of upward jumps;

(ii) a newly acquired task remains feasible for at least H unless a jump occurs.

An aircraft whose protection level is ρ can hold a task with alert limit α under this rule only if $\Delta t \le (\alpha - \rho)/L_\rho$.

With the measured constants, an aircraft at the benign ceiling $\bar{\rho}_0 = 0.94\,\mathrm{m}$ can hold a 2 m corridor only if $\Delta t \le (2 - 0.94)/5.30 = 0.200\,\mathrm{s}$. This is the allocator rate of $5.01\,\mathrm{Hz}$ quoted in the abstract: the rate at which *every* benign aircraft remains usable for the most demanding task. Aircraft with a lower benign level tolerate slower rates. Section 5.6 tests the rule at 2 Hz and at 5.3 Hz. Jumps cannot be predicted from a rate bound: a stale correction arrives all at once. Theorem 3 therefore isolates exactly what prediction can and cannot remove.

Proposition 3 (Misreported certificates)**.** *If f aircraft report a protection level other than their true one, the reported deficiency differs from the true one by at most f. Moreover, every violation of (C) under Π is committed by a misreporting aircraft.*

The deficiency depends on the certificates only through the count profile (n_k), which needs one counter per alert-limit class. Counts can be agreed in a decentralised fleet by resilient consensus [50]. Authenticated certificate reports, whose cost is budgeted in [12], reduce f to the number of compromised signing keys.

3.4 From one operation to a population

Theorem 1 is a $\mathbf{Q}_{\mathrm{op}}$ statement: it holds surely for one operation, given Assumption 1. Fleet sizing and per-flight-hour objectives need $\mathbf{Q}_{\mathrm{pop}}$ statements, such as the probability that a fleet of size N covers every task at a random instant of a random operation. The companion systematisation shows that no chain of proofs alone reaches such an objective: every path crosses a statistical obligation [13]. The lift below names its own obligations.

Proposition 4 (Population lift)**.** *Group the alert limits into classes $\lambda_1 < \cdots < \lambda_c$ with m_ℓ tasks at or below λ_ℓ. Assume two obligations:*

O1 *at a random instant, and conditionally on the task set, the aircraft's classes* $\ell_i = \min\{\ell : \rho_i \leq \lambda_\ell\}$ *(with* $\ell_i = c + 1$*, "none", if* $\rho_i > \lambda_c$*) are independent across aircraft;*

O2 *the classes are identically distributed with law q, with q estimated from operations representative of deployment.*

Then

$$\mathbb{P}\big(d = 0\big) = \mathbb{P}\Big(\textstyle\sum_{\ell' \leq \ell} C_{\ell'} \geq m_\ell \ \ \forall \ell\Big),$$

where $C \sim \mathrm{Multinomial}(N, q)$ *over the* $c + 1$ *classes; this is computable exactly by dynamic programming in* $O(cN^2)$*. The smallest fleet* $N^\star(\delta)$ *with* $\mathbb{P}(d = 0) \geq 1 - \delta$ *is the fleet-sizing answer, and the deficiency tail* $\mathbb{P}(d \geq k)$ *follows the same way.*

O1 is precisely what a common-cause failure breaks. One jammer degrades every aircraft in its footprint at once. Section 5.7 shows that the independent model then *underestimates* multi-task loss, the operationally dangerous direction. Whenever zero violations are observed over n agent-epochs, we report the exact one-sided 95% Clopper–Pearson (CP) upper bound on the per-epoch violation rate [23]. It is approximately $3/n$, the "rule of three" [35] that also underlies the test-hour ladders of the Specific Operations Risk Assessment (SORA) [2, 13].

Remark 1 (Where the closed forms stop). Nestedness needs a scalar certificate against a scalar alert limit. With vector certificates (horizontal and vertical protection levels, link quality), eligibility becomes a dominance order. Maximum matching remains polynomial [37], but Equation (3) and the linear-time monitors are lost. Robust assignment under vertex failures is then hard to approximate in general [6]. The scalar case is therefore the largest one in which every monitor is a closed form.

4 Methodology II: Experimental Design

Certificate process. We generate $\rho_i(t)$ by (1) from three independent mechanisms per aircraft:

- dead reckoning under jamming or sensor faults: staleness grows at 1 s/s, Poisson onsets, exponential durations, and a 2 s re-anchoring delay;
- undetected relay-delay episodes over $H = 2$ hops: staleness jumps by the sum of two residuals drawn from the 1,346 measured undetected residuals;
- a benign floor drawn from the 10,621 measured benign residuals.

Spatial jammers (radius 200 m, random walk) add common-cause dead reckoning to every aircraft in their footprint. Rates are stated modelling assumptions and are swept; every magnitude is measured. Appendix B lists each constant with its provenance.

Environment. CERTSWARM is a vectorised environment of B fleets with $N = 12$ aircraft and $M = 8$ tasks on a 1×1 km arena. Aircraft fly at 15 m/s, and a task is on site within 30 m. Two spatial jammers roam the arena. Decision epochs are $\Delta t = 0.5$ s, and missions last 150 s. Alert limits are drawn from $\{2, 5, 10, 20\}$ m with values $\{4, 3, 2, 1\}$.

Each aircraft observes:

- its own position, ρ_i, the change in ρ_i, and whether it holds a task;
- for each task, in egocentric order of distance: relative position, alert limit, value, eligibility, and whether it or another aircraft holds the task;
- the fleet's count profile and the jammer offsets.

The jamming-blind variant zeroes every certificate-derived feature.

Learning. We use proximal policy optimisation (PPO) [68] in the multi-agent PPO (MAPPO) style [83]. A parameter-shared actor outputs, per aircraft, a categorical over {keep, idle, k-th nearest task}. A centralised critic sees the aircraft's observation and the fleet mean. Training uses 16 parallel fleets, rollouts of 150 epochs and 200 updates (480k fleet-epochs, 5.8×10^6 agent-epochs per run), with 5 training seeds per method. Appendix C lists all hyperparameters.

Methods.

- *Ours*: Algorithm 1 in the environment, reward r_{cert}, and deviation penalty $\lambda_{\mathrm{dev}} = 0.05$. The ablation sets $\lambda_{\mathrm{dev}} = 0$.
- *Masking*: invalid-action masking per aircraft [39], with conflicts resolved by the first claimant.
- *PPO-Lagrangian* [67, 74]: the certificate is observed but not enforced. The reward is nominal coverage minus λ times the violation rate; λ is updated by dual ascent to a limit of 0.5% violating agent-epochs.
- *Jamming-blind PPO*: no certificate features, nominal reward.
- Every learned policy without the layer is also evaluated *with the projection added at deployment*, separating what the policy learns from what the mechanism enforces.
- Non-learned references:
  - keep-or-nearest proposals with the projection;
  - the tightest-fit greedy allocator re-solved every epoch;
  - a fixed design margin that admits task j iff $\rho_0 + L_\rho \cdot 1\,\mathrm{s} \le \alpha_j$, i.e. one second of dead reckoning budgeted at design time;
  - keep-or-nearest with no certificate.

Metrics and statistics. We report:

- certified coverage r_{cert};
- the *coverage ratio*, feasible holders over the maximum $M - d(t)$ from Lemma 1, which separates "safe because idle" from "safe and maximal";
- violations per agent-epoch, counted per step and never averaged over episodes, in training and in evaluation;
- inter-epoch exposure and reassignments per agent-epoch.

Each learned method is evaluated greedily on 3 held-out seeds $\times$ 16 fleets per training seed. We report the interquartile mean (IQM) with 95% stratified-bootstrap confidence intervals (CIs) [7], and CP bounds whenever zero violations are observed.

5 Results

5.1 Correctness of the closed forms

Over 10,000 random nested instances (up to 10 aircraft and 10 tasks, with ties, continuous and heavy-tailed certificates), we compared every closed form with a brute-force oracle: Hopcroft–Karp matching [37], enumeration of all b-subsets, and enumeration of task subsets for value. The checks covered:

- Lemma 1, the greedy and lexicographic projections, and value optimality;
- Lemma 2 and Proposition 1;
- the coverage bound of Theorem 2.

The 9 checks total 83,872 comparisons with 0 disagreements. A further 4,000 instances per misreporting level confirm Proposition 3: no honest aircraft ever violated (C). This is a check on the mathematics and the implementation, not evidence about flight.

5.2 RQ1: violations during learning

Figure 2 tracks certified coverage and cumulative violations over training.

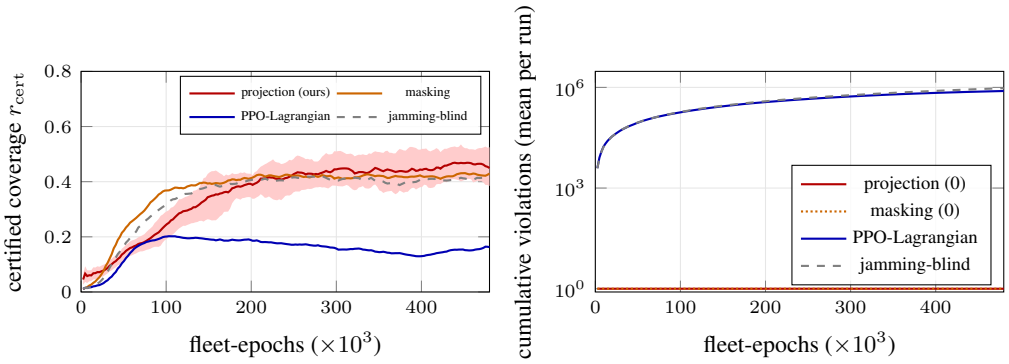


Figure 2: RQ1. Left: certified coverage during training (stochastic policy; mean over 5 seeds, 10-update moving average; band: min–max across seeds for ours). Right: cumulative executed violations on a log axis. Zero is drawn at 1 for visibility. The projection and masking curves lie on the floor throughout.

- *Projection in the environment.* It executed *zero* infeasible assignments over 2.9×10^7 agent-epochs, pooled across 5 seeds, from the first update onwards. Per-agent masking also reached zero, as separability predicts. For both methods zero holds by construction, so the CP bound in Table 2 only quantifies the exposure observed. That bound assumes independent agent-epochs, which these are not.
- *PPO-Lagrangian.* Although it observes the certificate, it executed 3,903,179 infeasible assignments while learning, 13.55% of agent-epochs. Its multiplier then rose until the policy largely stopped serving tasks. Final certified coverage is 0.036, and evaluation violations still average 0.73% of agent-epochs, above the 0.5% limit.
- *Jamming-blind PPO.* It executed 4,744,505 infeasible assignments (16.47%).

In-expectation methods violate the constraint while they learn, and learning is when a fleet would be flight-tested.

5.3 RQ3: what the policy learns and what the mechanism enforces

Table 2 evaluates every method greedily on held-out missions. Four findings stand out.

The mechanism does most of the work. Adding the projection at deployment to *any* learned policy removed every violation and lifted the coverage ratio to at least 0.99:

- jamming-blind: certified coverage $0.451 \rightarrow 0.492$;
- masking: $0.464 \rightarrow 0.499$;
- PPO-Lagrangian: $0.036 \rightarrow 0.393$.

Without it, the jamming-blind policy and the no-certificate heuristic reach seemingly competitive certified coverage (0.451 and 0.459). They do so only by holding tasks without certification on 12.67% and 13.97% of agent-epochs. The jamming-blind policy in fact converges to the keep-or-nearest rule, with no reassignments.

Masking is safe but not maximal. Its coverage ratio of 0.62 is the gap that Theorem 1(ii) closes. Agents that each pick an eligible task still collide.

Learning helps on top of the guarantee, but where the layer sits during training does not matter at this scale.

- The policy trained with the projection reaches 0.496 [0.473, 0.525] at a coverage ratio of 1.00.
- The keep-or-nearest heuristic with the same projection reaches 0.435, and the intervals barely overlap.
- Policies trained with masking, or blind, and projected at deployment do as well (0.499, 0.492).

Table 2: RQ3. Held-out evaluation ($N = 12$, $M = 8$; 5 training seeds $\times$ 3 evaluation seeds $\times$ 16 fleets for learned methods; 3 $\times$ 16 for the references below the line). Values are IQMs with 95% stratified-bootstrap CIs. Cov. ratio: feasible holders divided by $(M - d)$. Violations are in % of agent-epochs (IQM over runs). "0 ($\leq x$)": none observed, with the 95% CP upper bound on the rate, assuming independent agent-epochs. Train viol.: executed infeasible assignments over all of training.

Method	r_{cert}	cov. ratio	eval viol. [%]	train viol.	exposure [%]	reassign. [%]
Projection (SE) + PPO (ours)	0.496 [0.473, 0.525]	1.00	0 ($\leq 3.5 \times 10^{-6}$)	0 ($\leq 1.0 \times 10^{-7}$)	1.62	4.9
ablation: no deviation penalty	0.499 [0.484, 0.520]	1.00	0 ($\leq 3.5 \times 10^{-6}$)	0 ($\leq 1.0 \times 10^{-7}$)	1.66	4.9
Per-agent action masking + PPO	0.464 [0.458, 0.473]	0.62	0 ($\leq 3.5 \times 10^{-6}$)	0 ($\leq 1.0 \times 10^{-7}$)	0.52	0.7
+ projection at deployment	0.499 [0.489, 0.508]	1.00	0 ($\leq 3.5 \times 10^{-6}$)	–	1.71	4.9
PPO-Lagrangian (certificate observed)	0.036 [0.028, 0.047]	0.04	0.02	3,903,179	0.03	0.0
+ projection at deployment	0.393 [0.383, 0.405]	0.99	0 ($\leq 3.5 \times 10^{-6}$)	–	1.48	3.9
Jamming-blind PPO	0.451 [0.439, 0.467]	0.51	13.75	4,744,505	0.37	0.0
+ projection at deployment	0.492 [0.478, 0.508]	1.00	0 ($\leq 3.5 \times 10^{-6}$)	–	1.71	5.0
Keep-or-nearest + projection	0.435 [0.395, 0.477]	1.00	0 ($\leq 1.7 \times 10^{-5}$)	–	1.45	2.3
Tightest-fit greedy, re-solved each epoch	0.169 [0.142, 0.216]	0.98	0 ($\leq 1.7 \times 10^{-5}$)	–	1.99	8.6
Fixed design margin (1 s dead reckoning)	0.163 [0.155, 0.170]	0.31	8.03	–	0.20	0.0
Keep-or-nearest, no certificate	0.459 [0.433, 0.484]	0.51	13.97	–	0.38	0.0

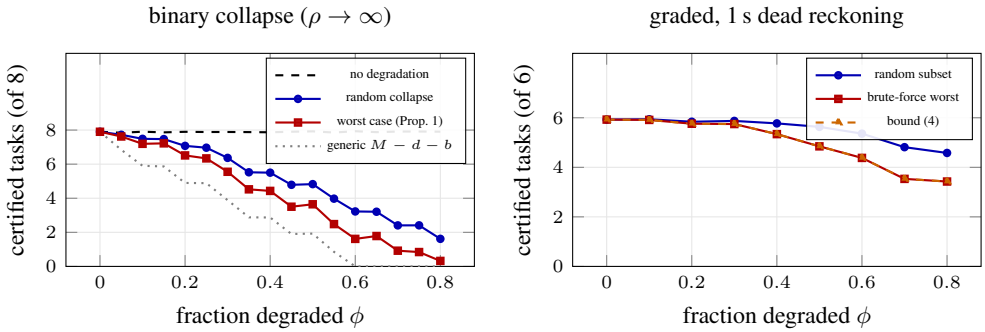


Figure 3: RQ2. Coverage as a fraction ϕ of the fleet degrades. Left: $N = 12$, $M = 8$, 400 fleets per point. Right: $N = 10$, $M = 6$, 150 fleets per point with exhaustive worst case; the bound of Proposition 2 lies on the worst case throughout.

- The deviation penalty makes no measurable difference (0.499 without it).

What training *with* the projection buys is the zero-violation learning of RQ1, not a better final policy.

Static and design-time allocators fail in opposite ways. The tightest-fit greedy re-solves from scratch every epoch, keeps its aircraft in transit (8.6% reassignments) and certifies only 0.169. A fixed design margin certifies 0.163 and still violates on 8.03% of agent-epochs, because a margin fixed at design time cannot track a certificate that moves.

5.4 RQ2: coverage as certificates degrade

Figure 3 shows fleets caught at a random instant of an operation, with heterogeneous certificates drawn from the stationary certificate process. Under binary collapse (left), the exact worst case of Proposition 1 lies well above the generic bound. At $\phi = 0.3$ ($b = 4$ of 12 aircraft), the fleet still certifies 5.55 tasks against the worst adversary and 6.37 under random collapse. The generic bound promises only 3.89. Under graded degradation of 1 s of dead reckoning (right), Proposition 2 coincided with the brute-force worst case in 100% of instances, with 0 violations of the bound.

5.5 RQ4: scalability

Figure 4 reports wall-clock latency on a two-core cloud CPU (x86-64). The closed-form monitor (deficiency and margin) takes 0.038 ms at $N = 1000$ and 6.2 ms at $N = 10^5$. The greedy nested projection takes 0.21 ms at $N = 1000$ and 40 ms at $N = 10^5$. The exact lexicographic projection

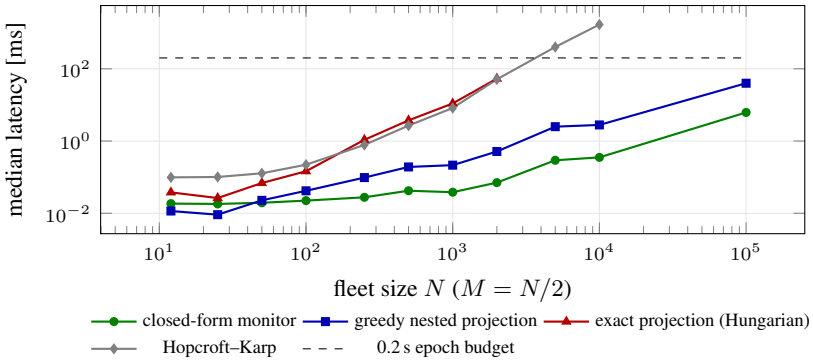


Figure 4: RQ4. Latency of the monitor and the projection layer against fleet size.

Table 3: RQ5. Projection rules between decision epochs (16 fleets $\times$ 4 seeds, 150 s missions). Exposure: holders feasible at an epoch but not before the next. "No jump": exposures with no upward certificate jump in the interval.

Rule	Δt	r_{cert}	cov. ratio	exposure [%]	exposures w/o jump
standard (C) only	0.5 s	0.393	0.983	1.51	2930
predictive release only	0.5 s	0.209	0.983	1.18	2182
predictive, $H = \Delta t$	0.5 s	0.263	0.834	0.19	0
predictive, $H = 1$ s	0.5 s	0.168	0.654	0.14	0
predictive, $H = 2$ s	0.5 s	0.083	0.377	0.10	0
standard (C) only	0.19 s	0.390	0.977	0.56	2842
predictive, $H = \Delta t$	0.19 s	0.371	0.983	0.09	0

(Hungarian) is the costly component: 0.038 ms for our 12-aircraft fleets, but 11.0 ms at $N = 1000$ and 53 ms at $N = 2000$. Generic Hopcroft–Karp matching takes 1682 ms at $N = 10^4$. Against the 0.200 s epoch of Theorem 3, the exact layer is real-time up to the largest size we timed ($N = 2000$), and the greedy layer beyond 10^5. Beyond the exact layer's range, the value and agreement parts of Theorem 1(ii) must be dropped or approximated; the cardinality guarantee is kept.

5.6 RQ5: between decision epochs

Table 3 runs the projection with keep-or-nearest proposals, isolating the allocation layer. At $\Delta t = 0.5$ s, the standard rule leaves 1.51% of agent-epochs exposed, 2930 of them without any certificate jump. The predictive rule of Theorem 3 cuts exposure by $8.1\times$ and leaves 0 exposures without a jump. In our simulator the drift between jumps equals L_ρ exactly and ρ is constant between epochs, so this zero is a sanity check of the implementation rather than evidence about flight. The rule also costs coverage, $0.393 \to 0.263$, because at 2 Hz the 2 m tasks become unholdable. Releasing early without predicting acquisitions ("release only") is worse: aircraft acquire, release and reacquire, and coverage drops to 0.209. At $\Delta t = 0.19$ s (5.3 Hz), just inside the rate of Theorem 3, the predictive rule costs little: coverage is 0.371 against 0.390. Exposure falls by $6.4\times$, and every remaining exposure is again caused by a jump. Hysteresis was much less useful here:

- the bound of Theorem 2 held for 100% of aircraft;
- a band of $\eta = 0.5$ reduced reassignments from 2.38% to 2.01% of agent-epochs, at a coverage cost of $0.393 \to 0.379$.

The reason is that the certificate moves in steps of $L_\rho \Delta t \approx 2.6$ m, which already exceed the bands of the demanding tasks. Chattering matters for smooth certificates; this one moves in steps.

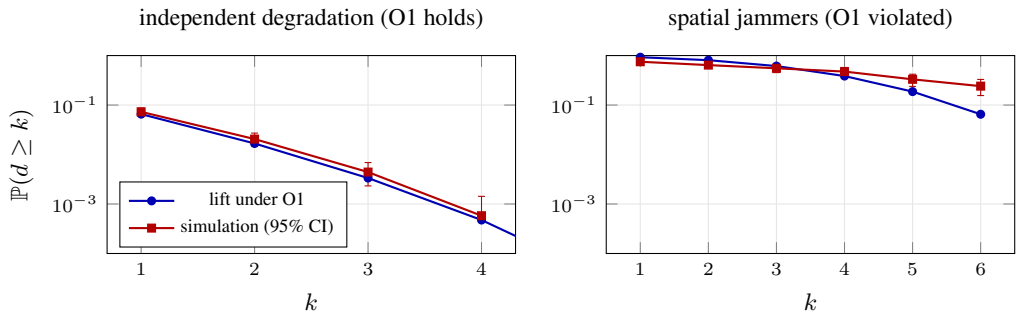


Figure 5: RQ6. Deficiency tail predicted by the population lift (Proposition 4) against simulation, $N = 12$, $M = 8$; CIs bootstrap over 64 independent fleets. Missing points are zero. The lift fails exactly where its independence obligation fails.

5.7 RQ6: the population lift and its obligation

We run 64 fleets ($N = 12$, $M = 8$, fixed alert limits) for 200 s, discard the first 25 s, and estimate the class law q from the remaining 175 s. We then compare the deficiency distribution that Proposition 4 predicts from q with the simulated one (Figure 5). Because q is estimated on the same runs, the marginals are matched by construction and only the dependence structure is tested.

- *Independent degradation, O1 true* (per-aircraft jamming onsets, no spatial jammers). The lift is accurate: $\mathbb{P}(d = 0)$ is 0.934 predicted against 0.927 simulated, and every tail agrees within its CI.
- *Common-cause jamming, O1 false.* Aircraft loiter around three operating areas while three jammers of radius 260 m drift at 4 m/s. At a random instant, 58% of aircraft are certified for no task. The independent model is now wrong in both directions:
  - it predicts full coverage 0.074 of the time against 0.250 [0.175, 0.334], too pessimistic by $3.4\times$;
  - it predicts that six or more of the eight tasks are lost 6.5% of the time against 24.1% [15.5, 32.9], too optimistic by $3.7\times$.

Correlation widens the distribution: jammers that miss a cluster leave it whole, and jammers that hit it take it all. A fleet sized by the independent model, $N^\star = 15$ aircraft for 99% full coverage under independent degradation, is therefore not sized for the catastrophic losses that a common-cause jammer produces. O1 must be evidenced by spatial separation or by a common-cause analysis, not assumed. This is the obligation-carrying lift argued for in [13], made quantitative.

6 Related Work

Constrained and safe RL. CMDPs [10] and their solvers are reviewed in [32, 77]:

- trust-region and primal-dual methods [4, 74, 81, 87], with zero duality gap [62] and near-optimal sample complexity [59];
- state-wise and high-probability formulations [84, 88, 89];
- almost-sure and anytime constraints [57, 58, 72];
- instantaneous hard constraints under unknown safety [70].

Anytime-constrained RL is the closest quantifier, and computing optimal policies there is NP-hard [57]. Its cost is generated by the agent's own trajectory, however. Ours is exogenous and is checked in closed form, which is why pointwise satisfaction is attainable without learning the constraint.

Safety layers, shields and filters. Safety layers [25, 63] and differentiable optimisation layers [8] project actions. Projection biases learning unless corrected [30], and the placement of the safeguard matters [56]. Krasowski et al. [46] survey the alternatives: action replacement, projection and masking. Shields provide zero-violation training in single- and multi-agent settings [9, 27, 80]. CBFs, including

graph-based ones for large swarms [11, 21, 65, 86], predictive safety filters [38, 76], reachability methods [15] and the Simplex architecture [69] protect a fixed safe set. They are complementary to this work. GCBF+ keeps aircraft apart; our layer decides which aircraft is certified for which task.

Safe MARL. MACPO and MAPPO-Lagrangian [31] bring trust-region and Lagrangian guarantees to cooperative MARL, and benchmark suites support evaluation [40, 41]. MAPPO [83], QMIX [66], MADDPG [53] and HAPPO [47] are the standard learners.

Certified and robust RL; exogenous action sets. Certified policies [48, 79, 82] and state-adversarial MDPs [51, 64, 85] certify a policy's response to a perturbation. Stochastic and changing action sets [18, 20, 44] and invalid-action masking [39] model exogenous availability. Online optimisation with adversarial constraints [55, 71] bounds cumulative violation. We need neither a distribution over availability nor cumulative slack: the constraint is known at decision time, and the difficulty lies in the *joint* (matching) structure and in its dynamics.

Matching and robust assignment. Hall's and Ore's theorems [33, 52, 61], difference and threshold graphs [34, 54], convex bipartite matching [29, 73], dynamic and weighted variants [19, 42], transversal matroids [26] and robust assignment [5, 6, 17, 36, 49] supply our tools. Our contribution is to bind them to a moving certificate and to a learning loop.

Multi-robot allocation and resilience. Our setting is single-task robots, single-robot tasks, instantaneous assignment [28, 45], with time-varying capability. Auction and consensus allocators [16, 22], learned schedulers [78] and capability-aware allocation [60] do not consume a formal certificate. Resilient submodular planning [75, 90, 91] bounds utility after binary worst-case removals, whereas Proposition 2 handles graded degradation exactly. GNSS spoofing threats [43] and integrity monitoring [1] give the aviation semantics, and SORA and European Union Aviation Safety Agency (EASA) guidance [2, 3] give the regulatory target.

7 Discussion, Limitations and Broader Impact

What the guarantee is. Theorem 1 is sure, not statistical, but it is conditional. It holds at decision epochs, given a sound certificate (Assumption 1). Theorem 3 extends it between epochs except across jumps, and the rate condition tells the designer how fast the allocator must run. Everything population-level rests on obligations O1–O2, which we show can fail.

Limitations.

- *Dependence on the certificate.* If the certificate is loose, constraints are conservative and coverage suffers. Feasibility and the bounds are unaffected.
- *Inherited caveats.* The parent certificate's own caveats carry over: a hover-regime calibration of γ, Lipschitz dynamics and estimator re-anchoring [14].
- *Simulation.* Our study is in simulation. The rates of the threat process are modelling assumptions; only its magnitudes are measured.
- *Scale of the learning study.* It is deliberately compact (12 aircraft, 5 seeds, CPU only). The guarantees hold for any policy, including a random one, so they do not depend on training scale. The comparative return figures do, and the runbook in our artifact scales them to larger fleets and to software-in-the-loop flight.
- *Scalar certificates only.* Vector certificates lose the closed forms (Remark 1).

Broader impact. The work aims at safer autonomous fleets in public airspace. UAV swarms are dual-use, and a method that keeps a fleet operating under jamming could serve military ends. Our contribution is an assurance mechanism that withholds tasks from aircraft that cannot be trusted with them; it does not increase any capability to cause harm. We release code and data under the artifact's licence.

8 Conclusion

Treating a formal certificate as the constraint set of a multi-agent policy changes what safe RL can promise. Because a scalar certificate makes eligibility nested, the coverage the fleet can certify, the failures it tolerates and the exact worst case under graded degradation are linear-time closed forms. A projection in the environment then makes every executed assignment feasible and coverage-maximal for any policy, during learning as during deployment. The remaining risk becomes visible as two quantities: exposure caused by certificate jumps, which no rate bound can predict, and the independence obligation of the population lift, which common-cause jamming violates. Both are what an airworthiness argument has to evidence next.

References

[1] Minimum operational performance standards for global positioning system/satellite-based augmentation system airborne equipment. Technical Report DO-229E, RTCA, Inc., 2016.

[2] JARUS guidelines on specific operations risk assessment (SORA). Technical Report JAR-DEL-WG6-D.04, Edition 2.0, Joint Authorities for Rulemaking of Unmanned Systems, 2019.

[3] EASA concept paper: Guidance for level 1 & 2 machine learning applications, issue 02. Technical report, European Union Aviation Safety Agency, 2024.

[4] Joshua Achiam, David Held, Aviv Tamar, and Pieter Abbeel. Constrained policy optimization. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, volume 70 of *PMLR*, pages 22–31, 2017.

[5] David Adjiashvili, Viktor Bindewald, and Dennis Michaels. Robust assignments via ear decompositions and randomized rounding. In *43rd International Colloquium on Automata, Languages, and Programming (ICALP)*, volume 55 of *LIPIcs*, pages 71:1–71:14, 2016.

[6] David Adjiashvili, Viktor Bindewald, and Dennis Michaels. Robust assignments with vulnerable nodes. *arXiv preprint arXiv:1703.06074*, 2017.

[7] Rishabh Agarwal, Max Schwarzer, Pablo Samuel Castro, Aaron Courville, and Marc G. Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, 2021.

[8] Akshay Agrawal, Brandon Amos, Shane Barratt, Stephen Boyd, Steven Diamond, and J. Zico Kolter. Differentiable convex optimization layers. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.

[9] Mohammed Alshiekh, Roderick Bloem, Rüdiger Ehlers, Bettina Könighofer, Scott Niekum, and Ufuk Topcu. Safe reinforcement learning via shielding. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, pages 2669–2678, 2018. doi: 10.1609/aaai.v32i1.11797.

[10] Eitan Altman. *Constrained Markov Decision Processes*. Chapman & Hall/CRC, 1999.

[11] Aaron D. Ames, Samuel Coogan, Magnus Egerstedt, Gennaro Notomista, Koushil Sreenath, and Paulo Tabuada. Control barrier functions: Theory and applications. In *18th European Control Conference (ECC)*, pages 3420–3431, 2019. doi: 10.23919/ECC.2019.8796030.

[12] Anonymous. What authenticity costs in metres: A certified budget for cryptography and sensing in UAV navigation, 2026. Companion paper, under review.

[13] Anonymous. SoK: The quantifier gap between certified robustness and airworthiness assurance, 2026. Companion paper, under review.

[14] Anonymous. From wire to flight: A conditional mission-completion guarantee for UAVs, 2026. Under review; artifact released with this submission.

[15] Somil Bansal, Mo Chen, Sylvia Herbert, and Claire J. Tomlin. Hamilton–Jacobi reachability: A brief overview and recent advances. In *IEEE 56th Conference on Decision and Control (CDC)*, 2017.

[16] Dimitri P. Bertsekas. The auction algorithm: A distributed relaxation method for the assignment problem. *Annals of Operations Research*, 14:105–123, 1988.

[17] Dimitris Bertsimas and Melvyn Sim. The price of robustness. *Operations Research*, 52(1):35–53, 2004.

[18] Craig Boutilier, Alon Cohen, Amit Daniely, Avinatan Hassidim, Yishay Mansour, Ofer Meshi, Martin Mladenov, and Dale Schuurmans. Planning and learning with stochastic action sets. In *Proceedings of the 27th International Joint Conference on Artificial Intelligence (IJCAI)*, pages 4674–4682, 2018. doi: 10.24963/ijcai.2018/650.

[19] Gerth Stølting Brodal, Loukas Georgiadis, Kristoffer Arnsfelt Hansen, and Irit Katriel. Dynamic matchings in convex bipartite graphs. In *Mathematical Foundations of Computer Science (MFCS)*, volume 4708 of *LNCS*, pages 406–417, 2007.

[20] Yash Chandak, Georgios Theocharous, Chris Nota, and Philip S. Thomas. Lifelong learning with a changing action set. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, 2020.

[21] Richard Cheng, Gábor Orosz, Richard M. Murray, and Joel W. Burdick. End-to-end safe reinforcement learning through barrier functions for safety-critical continuous control tasks. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, 2019.

[22] Han-Lim Choi, Luc Brunet, and Jonathan P. How. Consensus-based decentralized auctions for robust task allocation. *IEEE Transactions on Robotics*, 25(4):912–926, 2009. doi: 10.1109/TRO.2009.2022423.

[23] C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934.

[24] Jorge Cortés, Sonia Martínez, Timur Karataş, and Francesco Bullo. Coverage control for mobile sensing networks. *IEEE Transactions on Robotics and Automation*, 20(2):243–255, 2004.

[25] Gal Dalal, Krishnamurthy Dvijotham, Matej Vecerik, Todd Hester, Cosmin Paduraru, and Yuval Tassa. Safe exploration in continuous action spaces. *arXiv preprint arXiv:1801.08757*, 2018.

[26] Jack Edmonds and D. R. Fulkerson. Transversals and matroid partition. *Journal of Research of the National Bureau of Standards*, 69B:147–153, 1965.

[27] Ingy ElSayed-Aly, Suda Bharadwaj, Christopher Amato, Rüdiger Ehlers, Ufuk Topcu, and Lu Feng. Safe multi-agent reinforcement learning via shielding. In *Proceedings of the 20th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, pages 483–491, 2021.

[28] Brian P. Gerkey and Maja J. Matarić. A formal analysis and taxonomy of task allocation in multi-robot systems. *The International Journal of Robotics Research*, 23(9):939–954, 2004.

[29] Fred Glover. Maximum matching in a convex bipartite graph. *Naval Research Logistics Quarterly*, 14(3): 313–316, 1967.

[30] Sébastien Gros, Mario Zanon, and Alberto Bemporad. Safe reinforcement learning via projection on a safe set: How to achieve optimality? *IFAC-PapersOnLine*, 53(2):8076–8081, 2020.

[31] Shangding Gu, Jakub Grudzien Kuba, Yuanpei Chen, Yali Du, Long Yang, Alois Knoll, and Yaodong Yang. Safe multi-agent reinforcement learning for multi-robot control. *Artificial Intelligence*, 319:103905, 2023. doi: 10.1016/j.artint.2023.103905.

[32] Shangding Gu, Long Yang, Yali Du, Guang Chen, Florian Walter, Jun Wang, and Alois Knoll. A review of safe reinforcement learning: Methods, theories and applications. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2024.

[33] Philip Hall. On representatives of subsets. *Journal of the London Mathematical Society*, 10:26–30, 1935.

[34] Peter L. Hammer, Uri N. Peled, and Xiaorong Sun. Difference graphs. *Discrete Applied Mathematics*, 28 (1):35–44, 1990. doi: 10.1016/0166-218X(90)90092-Q.

[35] James A. Hanley and Abby Lippman-Hand. If nothing goes wrong, is everything all right? interpreting zero numerators. *JAMA*, 249(13):1743–1745, 1983.

[36] Felix Hommelsheim, Moritz Mühlenthaler, and Oliver Schaudt. How to secure matchings against edge failures. In *36th International Symposium on Theoretical Aspects of Computer Science (STACS)*, LIPIcs, 2019. doi: 10.4230/LIPIcs.STACS.2019.38.

[37] John E. Hopcroft and Richard M. Karp. An $n^{5/2}$ algorithm for maximum matchings in bipartite graphs. *SIAM Journal on Computing*, 2(4):225–231, 1973.

[38] Kai-Chieh Hsu, Haimin Hu, and Jaime F. Fisac. The safety filter: A unified view of safety-critical control in autonomous systems. *Annual Review of Control, Robotics, and Autonomous Systems*, 7:47–72, 2024. doi: 10.1146/annurev-control-071723-102940.

[39] Shengyi Huang and Santiago Ontañón. A closer look at invalid action masking in policy gradient algorithms. In *Proceedings of the International FLAIRS Conference*, volume 35, 2022. doi: 10.32473/flairs.v35i.130584.

[40] Jiaming Ji, Borong Zhang, Jiayi Zhou, Xuehai Pan, Weidong Huang, Ruiyang Sun, Yiran Geng, Yifan Zhong, Juntao Dai, and Yaodong Yang. Safety-gymnasium: A unified safe reinforcement learning benchmark. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023.

[41] Jiaming Ji, Jiayi Zhou, Borong Zhang, Juntao Dai, Xuehai Pan, Ruiyang Sun, Weidong Huang, Yiran Geng, Mickel Liu, and Yaodong Yang. OmniSafe: An infrastructure for accelerating safe reinforcement learning research. *Journal of Machine Learning Research*, 2024.

[42] Irit Katriel. Matchings in node-weighted convex bipartite graphs. *INFORMS Journal on Computing*, 20(2): 205–211, 2008. doi: 10.1287/ijoc.1070.0232.

[43] Andrew J. Kerns, Daniel P. Shepard, Jahshan A. Bhatti, and Todd E. Humphreys. Unmanned aircraft capture and control via GPS spoofing. *Journal of Field Robotics*, 31(4):617–636, 2014.

[44] Robert Kleinberg, Alexandru Niculescu-Mizil, and Yogeshwer Sharma. Regret bounds for sleeping experts and bandits. *Machine Learning*, 80:245–272, 2010.

[45] G. Ayorkor Korsah, Anthony Stentz, and M. Bernardine Dias. A comprehensive taxonomy for multi-robot task allocation. *The International Journal of Robotics Research*, 32(12):1495–1512, 2013.

[46] Hanna Krasowski, Jakob Thumm, Marlon Müller, Lukas Schäfer, Xiao Wang, and Matthias Althoff. Provably safe reinforcement learning: Conceptual analysis, survey, and benchmarking. *Transactions on Machine Learning Research*, 2023.

[47] Jakub Grudzien Kuba, Ruiqing Chen, Muning Wen, Ying Wen, Fanglei Sun, Jun Wang, and Yaodong Yang. Trust region policy optimisation in multi-agent reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2022.

[48] Aounon Kumar, Alexander Levine, and Soheil Feizi. Policy smoothing for provably robust reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2022.

[49] Pierre Laroche, Franc Marchetti, Sébastien Martin, and Zsuzsanna Róka. Bipartite complete matching vertex interdiction problem: Application to robust nurse assignment. In *International Conference on Control, Decision and Information Technologies (CoDIT)*, pages 182–187, 2014. doi: 10.1109/CoDIT.2014.6996890.

[50] Heath J. LeBlanc, Haotian Zhang, Xenofon Koutsoukos, and Shreyas Sundaram. Resilient asymptotic consensus in robust networks. *IEEE Journal on Selected Areas in Communications*, 31(4):766–781, 2013.

[51] Zuxin Liu, Zijian Guo, Zhepeng Cen, Huan Zhang, Jie Tan, Bo Li, and Ding Zhao. On the robustness of safe reinforcement learning under observational perturbations. In *International Conference on Learning Representations (ICLR)*, 2023.

[52] László Lovász and Michael D. Plummer. *Matching Theory*, volume 29 of *Annals of Discrete Mathematics*. North-Holland, 1986.

[53] Ryan Lowe, Yi Wu, Aviv Tamar, Jean Harb, Pieter Abbeel, and Igor Mordatch. Multi-agent actor-critic for mixed cooperative-competitive environments. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.

[54] N. V. R. Mahadev and Uri N. Peled. *Threshold Graphs and Related Topics*, volume 56 of *Annals of Discrete Mathematics*. North-Holland, 1995.

[55] Mehrdad Mahdavi, Rong Jin, and Tianbao Yang. Trading regret for efficiency: Online convex optimization with long term constraints. *Journal of Machine Learning Research*, 13:2503–2528, 2012.

[56] Hannah Markgraf, Shamburaj Sawant, Hanna Krasowski, Lukas Schäfer, Sébastien Gros, and Matthias Althoff. Safe reinforcement learning using action projection: Safeguard the policy or the environment? *Transactions on Machine Learning Research*, 2025. arXiv:2509.12833.

[57] Jeremy McMahan and Xiaojin Zhu. Anytime-constrained reinforcement learning. In *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 238 of *PMLR*, pages 4321–4329, 2024.

[58] Jeremy McMahan and Xiaojin Zhu. Anytime-constrained multi-agent reinforcement learning. *arXiv preprint arXiv:2410.23637*, 2024.

[59] Washim Uddin Mondal and Vaneet Aggarwal. Sample-efficient constrained reinforcement learning with general parameterization. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, pages 68380–68405, 2024.

[60] Gennaro Notomista, Siddharth Mayya, Seth Hutchinson, and Magnus Egerstedt. An optimal task allocation strategy for heterogeneous multi-robot systems. In *18th European Control Conference (ECC)*, 2019.

[61] Oystein Ore. Graphs and matching theorems. *Duke Mathematical Journal*, 22(4):625–639, 1955. doi: 10.1215/S0012-7094-55-02268-7.

[62] Santiago Paternain, Luiz F. O. Chamon, Miguel Calvo-Fullana, and Alejandro Ribeiro. Constrained reinforcement learning has zero duality gap. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.

[63] Tu-Hoa Pham, Giovanni De Magistris, and Ryuki Tachibana. OptLayer – practical constrained optimization for deep reinforcement learning in the real world. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2018.

[64] Lerrel Pinto, James Davidson, Rahul Sukthankar, and Abhinav Gupta. Robust adversarial reinforcement learning. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017.

[65] Zengyi Qin, Kaiqing Zhang, Yuxiao Chen, Jingkai Chen, and Chuchu Fan. Learning safe multi-agent control with decentralized neural barrier certificates. In *International Conference on Learning Representations (ICLR)*, 2021.

[66] Tabish Rashid, Mikayel Samvelyan, Christian Schroeder de Witt, Gregory Farquhar, Jakob Foerster, and Shimon Whiteson. Monotonic value function factorisation for deep multi-agent reinforcement learning. *Journal of Machine Learning Research*, 21(178):1–51, 2020.

[67] Alex Ray, Joshua Achiam, and Dario Amodei. Benchmarking safe exploration in deep reinforcement learning. Technical report, OpenAI, 2019.

[68] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

[69] Lui Sha. Using simplicity to control complexity. *IEEE Software*, 18(4):20–28, 2001.

[70] Ming Shi, Yingbin Liang, and Ness Shroff. A near-optimal algorithm for safe reinforcement learning under instantaneous hard constraints. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023.

[71] Abhishek Sinha and Rahul Vaze. Optimal algorithms for online convex optimization with adversarial constraints. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, pages 41274–41302, 2024.

[72] Aivar Sootla, Alexander I. Cowen-Rivers, Taher Jafferjee, Ziyan Wang, David H. Mguni, Jun Wang, and Haitham Bou Ammar. Sauté RL: Almost surely safe reinforcement learning using state augmentation. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, volume 162 of *PMLR*, pages 20423–20443, 2022.

[73] George Steiner and Julian Scott Yeomans. A linear time algorithm for maximum matchings in convex, bipartite graphs. *Computers & Mathematics with Applications*, 31(12):91–96, 1996.

[74] Adam Stooke, Joshua Achiam, and Pieter Abbeel. Responsive safety in reinforcement learning by PID Lagrangian methods. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, volume 119 of *PMLR*, pages 9133–9143, 2020.

[75] Vasileios Tzoumas, Konstantinos Gatsis, Ali Jadbabaie, and George J. Pappas. Resilient monotone submodular function maximization. In *IEEE 56th Conference on Decision and Control (CDC)*, 2017.

[76] Kim P. Wabersich and Melanie N. Zeilinger. A predictive safety filter for learning-based control of constrained nonlinear dynamical systems. *Automatica*, 129:109597, 2021. doi: 10.1016/j.automatica.2021.109597.

[77] Akifumi Wachi, Xun Shen, and Yanan Sui. A survey of constraint formulations in safe reinforcement learning. In *Proceedings of the 33rd International Joint Conference on Artificial Intelligence (IJCAI)*, 2024. doi: 10.24963/ijcai.2024/913.

[78] Zheyuan Wang and Matthew Gombolay. Learning scheduling policies for multi-robot coordination with graph attention networks. *IEEE Robotics and Automation Letters*, 5(3):4509–4516, 2020.

[79] Fan Wu, Linyi Li, Zijian Huang, Yevgeniy Vorobeychik, Ding Zhao, and Bo Li. CROP: Certifying robust policies for reinforcement learning through functional smoothing. In *International Conference on Learning Representations (ICLR)*, 2022.

[80] Wenli Xiao, Yiwei Lyu, and John M. Dolan. Model-based dynamic shielding for safe and efficient multi-agent reinforcement learning. *arXiv preprint arXiv:2304.06281*, 2023.

[81] Tengyu Xu, Yingbin Liang, and Guanghui Lan. CRPO: A new approach for safe reinforcement learning with convergence guarantee. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *PMLR*, 2021.

[82] Chenxi Yang, Greg Anderson, and Swarat Chaudhuri. CAROL: Certifiably robust reinforcement learning through model-based abstract interpretation. In *IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*, 2024.

[83] Chao Yu, Akash Velu, Eugene Vinitsky, Jiaxuan Gao, Yu Wang, Alexandre Bayen, and Yi Wu. The surprising effectiveness of PPO in cooperative multi-agent games. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2022.

[84] Dongjie Yu, Haitong Ma, Shengbo Li, and Jianyu Chen. Reachability constrained reinforcement learning. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, volume 162 of *PMLR*, pages 25636–25655, 2022.

[85] Huan Zhang, Hongge Chen, Chaowei Xiao, Bo Li, Mingyan Liu, Duane Boning, and Cho-Jui Hsieh. Robust deep reinforcement learning against adversarial perturbations on state observations. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, 2020.

[86] Songyuan Zhang, Oswin So, Kunal Garg, and Chuchu Fan. GCBF+: A neural graph control barrier function framework for distributed safe multiagent control. *IEEE Transactions on Robotics*, 41:1533–1552, 2025. doi: 10.1109/TRO.2025.3530348.

[87] Yiming Zhang, Quan Vuong, and Keith W. Ross. First order constrained optimization in policy space. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, 2020.

[88] Weiye Zhao, Tairan He, Rui Chen, Tianhao Wei, and Changliu Liu. State-wise safe reinforcement learning: A survey. In *Proceedings of the 32nd International Joint Conference on Artificial Intelligence (IJCAI)*, pages 6814–6822, 2023. doi: 10.24963/ijcai.2023/763.

[89] Weiye Zhao, Rui Chen, Yifan Sun, Tianhao Wei, and Changliu Liu. State-wise constrained policy optimization. *Transactions on Machine Learning Research*, 2024.

[90] Lifeng Zhou, Vasileios Tzoumas, George J. Pappas, and Pratap Tokekar. Resilient active target tracking with multiple robots. *IEEE Robotics and Automation Letters*, 4(1):129–136, 2019.

[91] Lifeng Zhou, Vasileios Tzoumas, George J. Pappas, and Pratap Tokekar. Distributed attack-robust submodular maximization for multirobot planning. *IEEE Transactions on Robotics*, 38(5):3097–3112, 2022.

A Proofs

Proof of Lemma 1. Ore's theorem [61] gives the maximum matching size in a bipartite graph as $M - \max_{S\subseteq\mathcal{T}}(|S| - |N(S)|)$. In a nested graph, the neighbourhood of a task set S equals the neighbourhood of its least demanding member, say task (k). Hence $|N(S)| = n_k$, while $|S| \le k$ with equality for $S = \{(1), \dots, (k)\}$. The maximum over S is therefore attained on prefixes and equals (3). The count profile needs one sort of ρ and one of α, plus M binary searches.

For the greedy rule, process tasks in ascending α. When task (k) is reached, every still-free eligible aircraft is also eligible for every later task, so the choice among them does not affect later feasibility. Let u_k be the number of tasks among the first k that the greedy assigns. A failure at (k) means that all n_k eligible aircraft are already used, so $u_{k-1} = n_k$ and $u_k = n_k$. Let $k^\star$ be the last failure. Then $u_{k^\star} = n_{k^\star}$, and every later task succeeds. The number of failures is $k^\star - n_{k^\star} + (\text{later failures} = 0) = k^\star - n_{k^\star} \le d$. It is also at least d, since no matching beats (3). Hence the greedy is maximum. □

Proof of Lemma 2. Removing a set R of b aircraft lowers n_k by $|R \cap \{i : \rho_i \le \alpha_{(k)}\}| \le b$. The bound is attained for every k simultaneously by removing the b smallest-ρ aircraft, which belong to every prefix set. By Lemma 1, coverage stays full iff $k \le n_k - b$ for all k. □

Proof of Proposition 1. After removal of the b smallest-ρ aircraft, the profile becomes $(n_k - b)^+$ for every k simultaneously. Any other set of b removals lowers each n_k by at most $\min(b, n_k)$. Apply Lemma 1. □

Proof of Proposition 2. Bound. Fix k. A degraded aircraft leaves the k-th prefix only if $\rho_i \le \alpha_{(k)} < \rho_i + s$. Aircraft with $\rho_i + s \le \alpha_{(k)}$ stay whatever happens. Hence the degraded profile satisfies $n'_k \ge \max\{n_k - b, n_k^{(s)}\}$, and Lemma 1 gives (4).

Attainment. Let $k^\star$ maximise the right-hand side. The band $\{i : \alpha_{(k^\star)} - s < \rho_i \le \alpha_{(k^\star)}\}$ has $n_{k^\star} - n_{k^\star}^{(s)}$ members. If b does not exceed that number, degrade any b of them, so $n'_{k^\star} = n_{k^\star} - b$. Otherwise degrade all of them, so $n'_{k^\star} = n_{k^\star}^{(s)}$. Then $d' \ge k^\star - n'_{k^\star}$, which equals the bound. □

Proof of Theorem 1. (i) Π returns a matching of E, and with zero bands E is exactly $G(t_k)$. This holds whatever the proposal, hence for every θ, at every epoch.

(ii) Every edge of E carries weight at least B. The value and agreement terms of a whole matching total at most $M(c_v \max v + 1) < B$. A maximum-weight matching therefore first maximises the number of matched pairs, which is $M - d$ by Lemma 1. For value, the task sets of maximum matchings are the bases of a transversal matroid [26]. If the task set of a maximum-weight matching σ did not have maximum value, the basis exchange property would give a basis with one task replaced and value larger by $v_j - v_{j'} \ge \min \Delta v$. Re-matching it changes agreement by at most

$\min(N, M) \le N$. The weight would then grow by at least $c_v \min \Delta v - N = 1 > 0$, contradicting maximality. Among maximum-value maximum matchings, the agreement term is all that remains.

(iii) Π is a deterministic function of (a, s) when s contains σ^-. If the threat process is Markov in s, the environment with Π inside it is therefore an MDP over proposals with kernel P_{Π}. The likelihood-ratio gradient is unbiased for any differentiable stochastic policy, including decentralised policies on partial observations, because it differentiates only the policy's log-probabilities. Since Π takes finitely many values, it is piecewise constant in any continuous relaxation of a, so its derivative vanishes almost everywhere.

The greedy variant inherits (i) and the cardinality part of (ii) from Lemma 1. □

Proof of Theorem 2. A forced release of task j by aircraft i at time t'' is preceded by an acquisition of j at some $t' < t''$, unless j was held at time 0, which happens at most once. At t', $\rho_i(t') \le (1-\eta)\alpha_j - h$. At t'', $\rho_i(t'') > \alpha_j$. The increase on $[t', t'']$ therefore exceeds $\eta\alpha_j + h \ge \eta\alpha_{(1)} + h$. Holding intervals of one aircraft are disjoint, so their increases add up to at most TV_i^+.

For coverage, every edge of the nested graph built from $\tilde{\rho}$ is an acquisition edge of E. It is also a keep edge whenever $\ell_{\mathrm{keep}} \le \ell_{\mathrm{acq}} + h$, since then $\rho_i + \ell_{\mathrm{acq}} \le (1-\eta)\alpha_j - h$ implies $\rho_i + \ell_{\mathrm{keep}} \le \alpha_j$. Hence E contains that nested graph, and its maximum matching is at least $M - d(\tilde{\rho})$. □

Proof of Theorem 3. Let (i, j) be executed at t_k. If j was kept, $\rho_i(t_k) + L_\rho \Delta t \le \alpha_j$. If j was acquired, $\rho_i(t_k) + L_\rho H \le \alpha_j$. Without an upward jump, $\rho_i(t) \le \rho_i(t_k) + L_\rho(t - t_k)$, which proves (i) on $[t_k, t_{k+1})$ and (ii) on $[t_k, t_k + H]$. For the rate condition, keeping requires $\rho + L_\rho \Delta t \le \alpha$. □

Proof of Proposition 3. Each misreport moves each n_k by at most one, so f misreports move each term of (3) by at most f. Π is feasible with respect to the reported values, and for honest aircraft the reported value is the true one. □

Proof of Proposition 4. With alert limits in classes, the prefix conditions of Lemma 1 bind only at class boundaries, so $d = 0$ iff $\sum_{\ell' \le \ell} C_{\ell'} \ge m_\ell$ for all ℓ, where C_ℓ counts aircraft of class ℓ. Under O1–O2, $C \sim \mathrm{Multinomial}(N, q)$. Sequential binomial splitting gives a dynamic programme over the cumulative count with $O(N)$ states and $O(N)$ transitions per class. □

B Constants and Provenance

Table 4: Constants. "Measured": computed from a released capture; "derived": arithmetic on measured values; "assumption": stated and swept.

Quantity	Value	Provenance	Source
γ_{sup}	1.625 m/s	measured	674 GNSS jamming/spoofing samples
L	1.181	measured	local Lipschitz estimate
e^{LT}, $T = 1$ s	3.26	derived	–
$L_\rho = e^{LT}\gamma_{\mathrm{sup}}$	5.30 m/s	derived	–
benign residual max	0.178 s	measured	10,621 benign hops
undetected residual (median / max)	3.94 / 4.84 s	measured	1,346 undetected hops
ρ_0 (benign floor)	0.94 m	derived	–
two-hop jump (median)	41.8 m	derived	–
alert limits	$\{2, 5, 10, 20\}$ m	assumption	parent corridor family
onset rates, durations	see code	assumption	swept

C Hyperparameters

PPO with clip 0.2, discount 0.995, generalised advantage estimation $\lambda = 0.95$, Adam with learning rate 3×10^{-4}, 4 epochs of 4 minibatches, entropy coefficient 0.003 and gradient-norm clip 0.5. Actor

and critic each have two hidden layers of 128 tanh units. The actor's keep logit is initialised at $+3$, so that an untrained allocator mostly keeps its tasks; all methods share this prior. For PPO-Lagrangian, the dual step is 1.0 per update and the limit is 0.005 violations per agent-epoch. Deviation penalty: $\lambda_{\mathrm{dev}} = 0.05$. A run takes about 5 minutes on one CPU core.

D Acronyms

CBF	control barrier function
CPO	constrained policy optimisation
PID	proportional–integral–derivative
CI	confidence interval
CMDP	constrained Markov decision process
CP	Clopper–Pearson
EASA	European Union Aviation Safety Agency
GNSS	global navigation satellite system
IQM	interquartile mean
GPS	Global Positioning System
MAPPO	multi-agent PPO
MACPO	multi-agent CPO
SCPO	state-wise CPO
MARL	multi-agent reinforcement learning
MDP	Markov decision process
PPO	proximal policy optimisation
RL	reinforcement learning
RQ	research question
SE	safeguard in environment
SORA	Specific Operations Risk Assessment
UAV	unmanned aerial vehicle